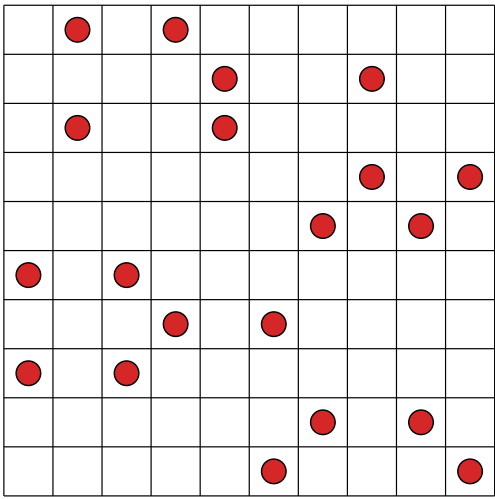


Four Methods, One Problem: The No-Three-In-Line Problem

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ILP certifies. PatternBoost learns. PPO places. SciML reaches.



Optimal 10×10 solution (20 pts)

- **ILP** certifies provably optimal solutions to $n = 19$ — no ML method matches this guarantee
- **SciML** (gradient flow, Apple M2 Metal, no symmetry pruning) solves $n = 18$ in 9/10 runs
- **Continuous relaxation** extends ML reach well beyond the discrete-search ceilings of transformer and RL methods

1. The No-Three-In-Line Problem

Place the **maximum** number of points on an $n \times n$ integer grid such that **no three points are collinear**. Trivial upper bound: $2n$ (pigeonhole). Best proven lower bound: $\approx 1.5n$ (Hall et al. 1975). Exact optimal known to $n \approx 19$ with symmetry pruning; $n \geq 20$ open.

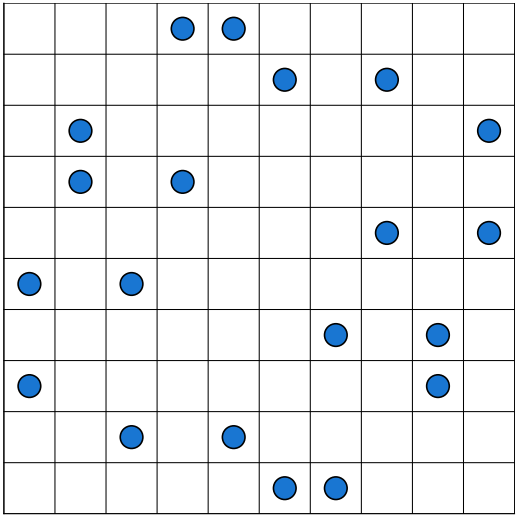
2a. Integer Linear Programming (ILP)

- Collinearity encoded as linear constraints; solved by Gurobi branch-and-bound
- **Certiably optimal** at every n tested ($n \leq 19$)
- Exponential worst-case scaling; symmetry pruning essential for $n \geq 15$

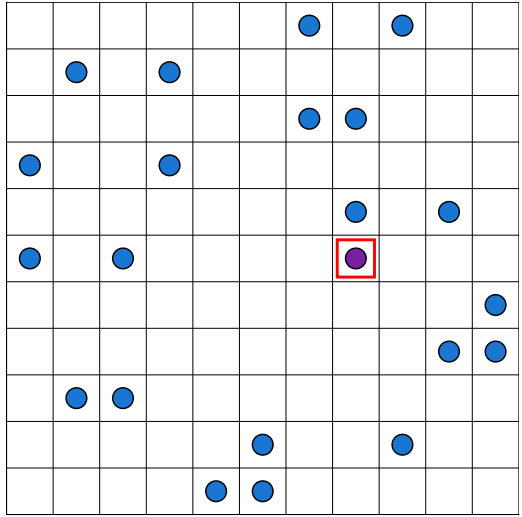
Certified optimal to $n = 19$

2c. Proximal Policy Optimisation (PPO)

- RL agent learns a point-placement policy on the $n \times n$ grid
- Perfect solutions at $n = 10$; one collinearity violation at $n = 11$ (shown right →)
- Discrete action space scales as n^2 — key scalability bottleneck



$n = 10$: 20 pts, 0 violations

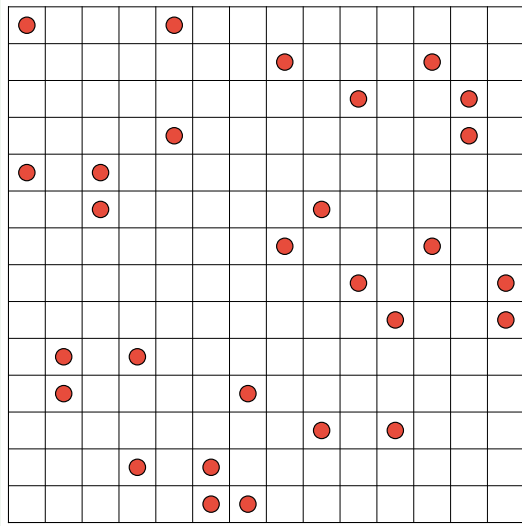


$n = 11$: 22 pts, 1 violation ←

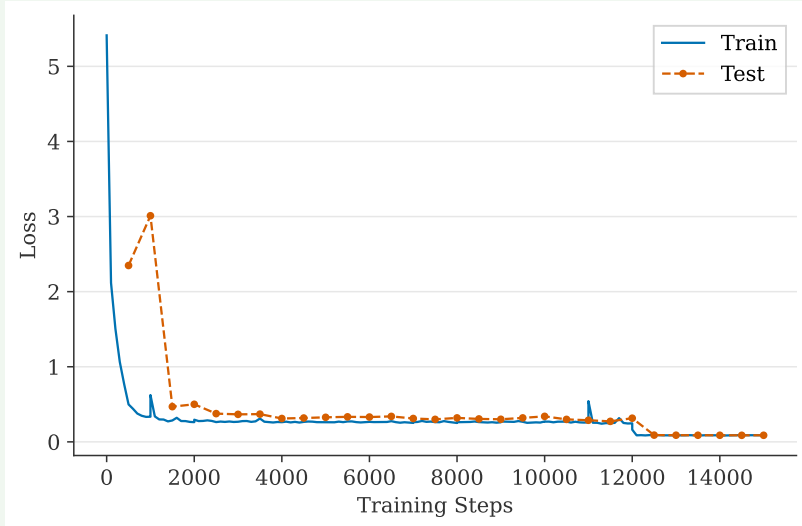
Hits constraint wall at $n = 11$

2b. PatternBoost (Transformer)

- Transformer trained on known solutions learns N3L geometric patterns
- Iterative boosting: weak candidates regenerated; matches optimal to $n = 14$
- Finds 29/30 points at $n = 15$ (optimal is 30)



$n = 14$: 28 pts (optimal)

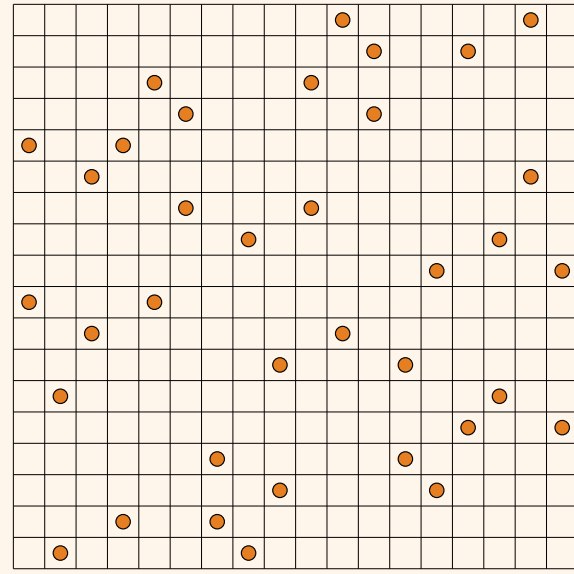


Training loss

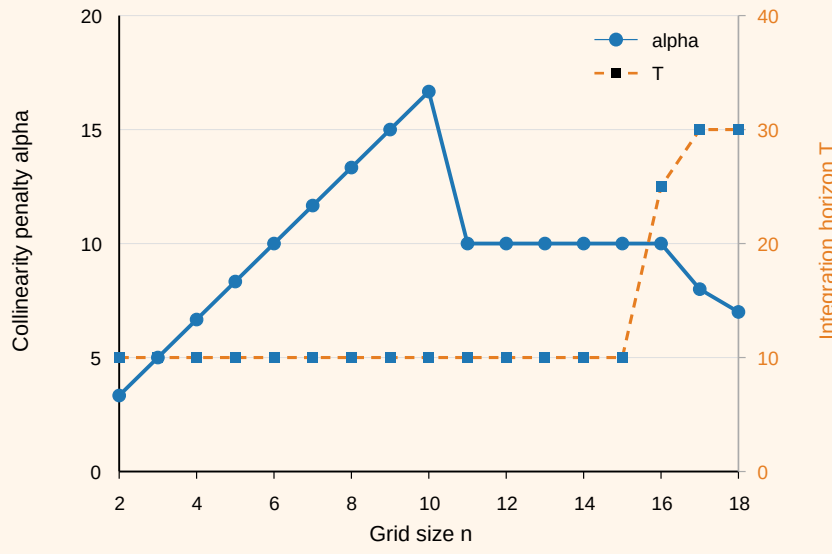
Optimal to $n = 14$; 29/30 at $n = 15$

2d. Scientific Machine Learning (SciML)

- Recasts N3L as continuous energy minimisation solved by gradient flow:
 $E = \alpha E_{\text{coll}} + E_{\text{count}} + \mu E_{\text{bin}}$
- Custom Metal GPU kernel (Julia, Apple M2); no symmetry pruning
- Non-monotonic reliability: $n = 17$ fails (0/10); $n = 18$ succeeds (9/10)



$n = 18$: 36 pts, 0 violations (9/10 runs)



α, T schedule across n

Reaches $n = 18$ without symmetry pruning

3. Results — Maximum Points Found (optimal = $2n$)

n	ILP	Pattern-Boost	PPO	SciML (10 runs)
10	20	20	20	20 (10/10)
11	22	22	22*	22 (7/10)
12	24	24	—	24 (10/10)
13	26	26	—	26 (8/10)
14	28	28	—	28 (4/10)
15	30	29	—	30 (8/10)
16	32	—	—	32 (8/10)
17	34	—	—	0/10 [†]
18	36	—	—	36 (9/10)
19	38	—	—	0/10

Optimal ($2n$) Sub-optimal Violation/failure Not tested

*PPO $n = 11$: 22 pts but 1 collinearity violation. [†]SciML $n = 17$: 0/10 standard seeds; optimal for one isolated seed only.

4. Takeaways

- **Continuous relaxation is the key advance:** SciML reaches $n = 18$ without symmetry pruning — beyond any prior ML solver for this problem
- **Exact guarantees still require ILP:** no learning-based method certifies optimality; the two paradigms are complementary, not competing
- **Non-monotonicity is informative:** $n = 17$ failure despite $n = 18$ success suggests energy-landscape sensitivity, not purely grid-size scaling
- **Future:** ILP+SciML warm-starting; extend to $n \geq 20$; investigate $n = 17$ failure

Paper & Code

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